

COURSE SYLLABUS

Introduction to SQL Databases (2025)

COURSE NUMBER

65315A

DURATION

3 days · 9 modules

AUDIENCE LEVEL

Beginner

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This 3-day instructor-led course provides an introductory foundation for anyone getting started with databases, including programmers and IT professionals expanding into database management. Students will learn fundamental database concepts through demonstrations and hands-on labs using SQL Server 2025 and SQL Server Management Studio, exploring key features and improvements in the latest version.

AT COURSE COMPLETION

After completing this course, you will be able to:

Describe fundamental database concepts, relational model principles, and SQL Server 2025 architecture components

Write T-SQL statements using DDL, DML, and DCL to create, modify, and query database objects while applying best practices

Design efficient database schemas using entity-relationship modeling, normalization techniques, and relationship implementation

Optimize database performance through indexing strategies, execution plan analysis, and appropriate use of database objects

Automate SQL Server administration tasks using PowerShell and leverage new SQL Server 2025 features including vector databases and enhanced temporal tables

PREREQUISITES

- Basic understanding of Windows operating system
 - Familiarity with general IT concepts
 - No prior database experience required
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COURSE OUTLINE

MODULE 1 Introduction to Databases

120 min

This module introduces fundamental database concepts, the relational model, and SQL Server 2025 architecture. Students will learn about database management systems, relational database principles, and the components of SQL Server.

LESSONS

LAB EXERCISES

Lesson 1: Understanding Databases and Database Management Systems

Exploring SQL Server 2025 Architecture and Management Studio

Lesson 2: The Relational Database Model

Lesson 3: SQL Server Editions and Versions

Lesson 4: SQL Server 2025 Architecture Overview

Lesson 5: System Databases in SQL Server

Lesson 6: SQL Server Management Studio (SSMS)

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Describe the purpose and characteristics of database management systems

Explain the relational model and its core principles including tables, rows, and columns

Identify SQL Server 2025 architecture components including instances, databases, and system databases

Navigate SQL Server Management Studio to connect to instances and explore database objects

Differentiate between system databases and user databases in SQL Server

This module introduces SQL and T-SQL, covering Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL). Students will learn to write and execute T-SQL statements following best practices.

LESSONS

Lesson 1: Introduction to SQL and T-SQL

Lesson 2: SQL Language Categories: DDL, DML, and DCL

Lesson 3: T-SQL Syntax Rules and Conventions

Lesson 4: Data Definition Language (DDL) in Practice

Lesson 5: Data Manipulation Language (DML) Essentials

Lesson 6: Filtering and Sorting Query Results

LAB EXERCISES

Writing and Executing Basic SQL Statements

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Differentiate between DDL, DML, and DCL statements and their purposes

Write CREATE, ALTER, and DROP statements to manage database objects

Construct SELECT queries to retrieve data from tables using proper syntax

Execute INSERT, UPDATE, and DELETE statements to modify table data

Apply T-SQL best practices including formatting, commenting, and semicolon terminators

Use the WHERE clause to filter query results based on specific criteria

MODULE 3 Data Modelling Techniques

150 min

This module covers data modeling fundamentals, entity-relationship modeling, and translating conceptual designs into physical database schemas. Students will learn to create ERDs and implement them in SQL Server.

LESSONS

- Lesson 1:** Introduction to Data Modeling
- Lesson 2:** Entities and Attributes Fundamentals
- Lesson 3:** Primary Keys and Candidate Keys
- Lesson 4:** Foreign Keys and Referential Integrity
- Lesson 5:** Entity-Relationship Diagrams and Notation
- Lesson 6:** Translating ERDs to Database Schemas

LAB EXERCISES

- Creating an Entity-Relationship Model for a Library System

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Explain the purpose and benefits of data modeling in database design
- Identify entities, attributes, and relationships in business scenarios
- Create entity-relationship diagrams using standard notation and symbols
- Define primary keys, foreign keys, and candidate keys for entities
- Translate conceptual ERD models into physical database table structures
- Apply entity and attribute naming conventions following best practices

This module teaches normalization theory and techniques, covering first through third normal forms and BCNF. Students will learn to normalize database designs to eliminate redundancy and understand when denormalization is appropriate.

LESSONS

- Lesson 1:** Introduction to Normalization
- Lesson 2:** Functional Dependencies and Data Anomalies
- Lesson 3:** First Normal Form (1NF)
- Lesson 4:** Second Normal Form (2NF)
- Lesson 5:** Third Normal Form (3NF)
- Lesson 6:** Boyce-Codd Normal Form and Higher Normal Forms
- Lesson 7:** Denormalization: Strategic Trade-offs

LAB EXERCISES

- Normalizing and Denormalizing Database Tables

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Explain the purpose and benefits of database normalization
- Identify functional dependencies and determinants in table structures
- Apply first normal form (1NF) rules to eliminate repeating groups
- Transform tables to second normal form (2NF) by removing partial dependencies
- Normalize tables to third normal form (3NF) by eliminating transitive dependencies
- Describe Boyce-Codd normal form (BCNF) and its application scenarios
- Evaluate when denormalization is appropriate for performance optimization

MODULE 5 Relationship Types and Effects in Database Design

140 min

This module explores relationship types in database design, including one-to-one, one-to-many, and many-to-many relationships. Students will learn to implement these relationships using foreign keys and junction tables.

LESSONS

Lesson 1: Understanding Relationship Fundamentals and Cardinality

Lesson 2: One-to-Many Relationships and Foreign Keys

Lesson 3: Many-to-Many Relationships and Junction Tables

Lesson 4: Composite Primary Keys in Junction Tables

Lesson 5: One-to-One Relationships and Implementation Patterns

Lesson 6: Referential Integrity and Cascade Options

LAB EXERCISES

Implementing Database Relationships and Referential Integrity

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Differentiate between one-to-one, one-to-many, and many-to-many relationships

Implement one-to-many relationships using foreign key constraints

Create junction tables to represent many-to-many relationships

Establish one-to-one relationships using appropriate constraint mechanisms

Configure referential integrity options including CASCADE, SET NULL, and NO ACTION

Analyze business rules to determine appropriate relationship types

Diagram relationships using ERD notation

This module examines how database design decisions impact performance, covering indexing strategies, execution plans, and performance optimization techniques. Students will learn to analyze and improve query performance.

LESSONS

- Lesson 1:** Database Design and Performance Relationship
- Lesson 2:** Index Fundamentals and Purposes
- Lesson 3:** Clustered Indexes and Table Structure
- Lesson 4:** Non-Clustered Indexes and Covering Indexes
- Lesson 5:** Index Selection Criteria and Strategies
- Lesson 6:** Execution Plans and Query Optimization
- Lesson 7:** Missing Index Recommendations and Statistics
- Lesson 8:** Index Fragmentation and Maintenance
- Lesson 9:** Indexing Best Practices and Common Pitfalls

LAB EXERCISES

- Optimizing Query Performance with Indexes

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Explain how database design affects query performance
- Describe clustered and non-clustered index structures and their differences
- Create appropriate indexes to optimize query performance
- Analyze execution plans to identify performance bottlenecks
- Interpret index usage statistics and recommendations
- Apply indexing best practices including column selection and index maintenance
- Identify over-indexing and its negative performance impacts

This module introduces essential database objects including views, stored procedures, functions, and triggers. Students will learn to create and use these objects to implement business logic and simplify data access.

LESSONS

Lesson 1: Database Objects Overview

Lesson 2: Views: Simplifying Data Access

Lesson 3: Stored Procedures Fundamentals

Lesson 4: User-Defined Functions

Lesson 5: Triggers for Automated Actions

Lesson 6: Choosing the Right Database Object

LAB EXERCISES

Creating and Using Views, Stored Procedures, Functions, and Triggers

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Create views to simplify complex queries and provide data abstraction

Develop stored procedures to encapsulate business logic and parameterized operations

Design user-defined functions for reusable calculations and transformations

Implement triggers to enforce business rules and audit data changes

Differentiate between scalar and table-valued functions

Apply best practices for stored procedure parameter handling and error management

Evaluate when to use each database object type based on requirements

MODULE 8 Introduction to PowerShell for SQL Server Administration

150 min

This module introduces PowerShell for SQL Server administration tasks, covering the SqlServer module, common cmdlets, and automation scenarios. Students will learn to script administrative operations using PowerShell.

LESSONS

Lesson 1: PowerShell Fundamentals for Database Administrators

Lesson 2: The SqlServer PowerShell Module

Lesson 3: Connecting to SQL Server with PowerShell

Lesson 4: Executing T-SQL with Invoke-Sqlcmd

Lesson 5: Automating Backup Operations with PowerShell

Lesson 6: Retrieving SQL Server Information with Get-Cmdlets

Lesson 7: PowerShell Script Structure and Best Practices

Lesson 8: Scheduling and Automating PowerShell Scripts

LAB EXERCISES

Automating SQL Server Administration with PowerShell

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Explain the benefits of using PowerShell for SQL Server administration

Install and import the SqlServer PowerShell module

Connect to SQL Server instances using PowerShell cmdlets

Execute T-SQL queries from PowerShell using Invoke-Sqlcmd

Automate database backup operations with PowerShell scripts

Retrieve SQL Server configuration and metadata using Get- cmdlets

Create reusable PowerShell scripts for common administrative tasks

This optional module explores new and improved features in SQL Server 2025 (version 17.x), including vector databases, enhanced temporal tables, and other significant updates. Students will gain awareness of modern SQL Server capabilities.

LESSONS

Lesson 1: SQL Server 2025 Overview and Release Highlights

Lesson 2: Vector Databases and AI Integration

Lesson 3: Enhanced Temporal Tables and Querying Capabilities

Lesson 4: Security Improvements and Compliance Features

Lesson 5: Performance Enhancements and Query Optimizer Updates

Lesson 6: Cloud Integration and Hybrid Scenarios

Lesson 7: JSON and XML Processing Improvements

Lesson 8: Feature Evolution and Migration Considerations

LAB EXERCISES

Exploring Vector Search and Enhanced Temporal Tables in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Identify new features introduced in SQL Server 2025

Describe vector database capabilities and use cases for AI workloads

Explain enhancements to temporal tables and system-versioned tables

Evaluate improvements in security, performance, and scalability

Demonstrate basic usage of new SQL Server 2025 features in labs

Compare SQL Server 2025 features with previous versions

BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3

SKILLS TO BE INTEGRATED

Connect to a SQL Server 2025 instance using SQL Server Management Studio and navigate the Object Explorer to view database components

Execute basic SELECT queries to retrieve data from existing tables using proper syntax and semicolon terminators

Create a new database using both GUI and T-SQL DDL commands with appropriate file and filegroup specifications

Create tables with appropriate data types, constraints, and column definitions using T-SQL DDL statements

Insert, update, and delete data in tables using DML statements with proper WHERE clause filtering

Design entity-relationship diagrams identifying entities, attributes, primary keys, and relationships using standard notation

Translate conceptual ERD designs into physical database schemas with appropriate table structures and column definitions

DAY 2 CAPSTONE Modules 4, 5, 6

SKILLS TO BE INTEGRATED

- Analyze table designs to identify functional dependencies and determine the current normal form
- Apply normalization rules to transform tables from lower normal forms (1NF, 2NF) to third normal form (3NF) by decomposing tables and eliminating redundancy
- Implement one-to-many relationships using foreign key constraints with proper referential integrity settings
- Create and implement many-to-many relationships using junction tables with composite primary keys
- Establish one-to-one relationships using unique constraints or shared primary keys as appropriate to the business scenario
- Create non-clustered indexes on frequently queried columns to improve SELECT statement performance
- Analyze query execution plans using SQL Server Management Studio to identify performance bottlenecks and missing index recommendations
- Implement clustered indexes on appropriate columns considering data distribution and query patterns

DAY 3 CAPSTONE Modules 7, 8, 9

SKILLS TO BE INTEGRATED

- Create views to encapsulate complex queries and provide simplified data access for users or applications
- Implement stored procedures with input parameters and output parameters to encapsulate business logic
- Design and create user-defined functions (scalar and table-valued) to perform reusable calculations or data transformations
- Create triggers (DML triggers) to enforce business rules or audit data changes automatically
- Connect to SQL Server instances using PowerShell and the SqlServer module cmdlets
- Execute T-SQL queries and scripts from PowerShell using Invoke-Sqlcmd with proper parameter handling
- Automate database backup operations using PowerShell scripts with Backup-SqlDatabase cmdlet
- Retrieve SQL Server configuration and database metadata using PowerShell Get- cmdlets for reporting or monitoring purposes

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.

